RECORDED CONTROL

A compact update core preserves nonzero singular values

Full updated factor

$[A, U]$

$\longleftrightarrow$

Compact update core

C_{core}

$$[A, U] = [Q, Q_{\perp}]C_{\mathrm{core}}$$

Maximum absolute discrepancy

4.263e-14

Relative spectrum discrepancy

5.890e-16

Recorded QR control: 128 rows, 180 initial columns, 37 appended columns.